Lecture 3: Classical computer vision

Hand-designed features: thresholding, Sobel, template matching

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University of Washington

August 20, 2026

Traditional Computer Vision

Core Techniques

- **Thresholding**
  Pixels $\geq$ a value $\rightarrow$ set to maximum
- Edge Detection
  Detect changes in brightness
- Segmentation
  Group regions enclosed by edges
- Curve Detection
  Link adjacent edges into shapes
- Optical Flow
  Estimate motion between frames

10.4018/978-1-5225-0140-4.ch004

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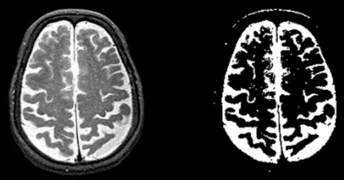

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Classical pipeline: Pixels $\rightarrow$ Edges $\rightarrow$ Shapes $\rightarrow$ Motion

Feature Engineering: Sobel Filters

Edge: a location where image intensity changes abruptly.

Intuition

- Flat region $\rightarrow$ small intensity change
- Object boundary $\rightarrow$ sharp intensity change

Treat the image as a function $f(x, y)$:

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

Edge strength (gradient magnitude):

$$\|\nabla f\| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

Goal: find pixels where the gradient magnitude is large.

Practical Significance

- Extract object boundaries
- Reduce data complexity

Numerical Method

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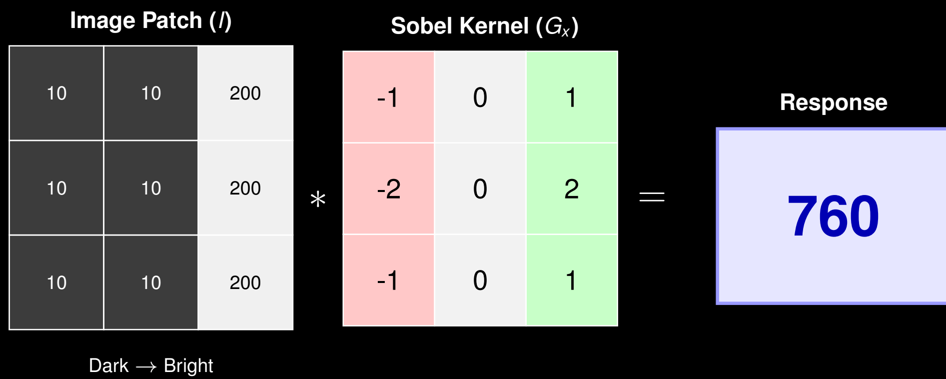
$\Rightarrow$ **Sobel filter**

Sobel Filter: Vertical Edge Detection

$$\underset{\text{Dark} \longrightarrow \text{Bright}}{\overset{\text{Image Patch } (I)}{\begin{bmatrix} 10 & 10 & 200 \\ 10 & 10 & 200 \\ 10 & 10 & 200 \end{bmatrix}}} * \overset{\text{Sobel Kernel } (G_x)}{\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}} = \overset{\text{Response}}{\boxed{760}}$$

Sobel Filter: Vertical Edge Detection



$$\text{Sum} = (-1 \cdot 10) + (0 \cdot 10) + (1 \cdot 200) + \cdots = \mathbf{760}$$

Sobel Filter: Vertical Edge Detection

Image Patch (I)

10	10	200
10	10	200
10	10	200

Dark $\longrightarrow$ Bright

$*$

Sobel Kernel (G_x)

-1	0	1
-2	0	2
-1	0	1

$=$

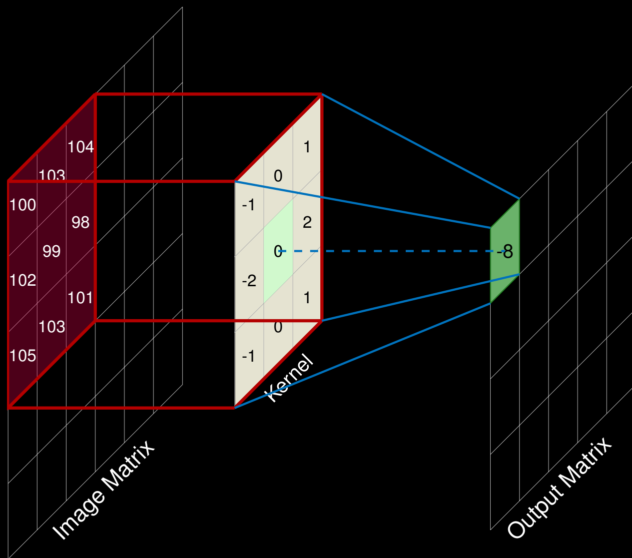
Response

760

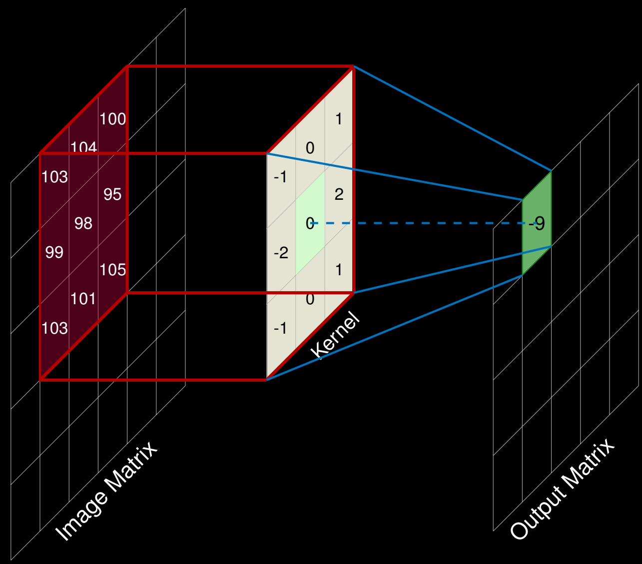
$$\text{Sum} = (-1 \cdot 10) + (0 \cdot 10) + (1 \cdot 200) + \cdots = \mathbf{760}$$

*The **negative weights** suppress dark pixels, while **positive weights** amplify bright pixels. A high response (760) indicates a strong **vertical edge**.*

Sobel Filter with Convolution

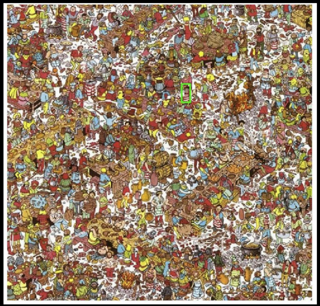


Sobel Filter with Convolution (Slide Right + 1)

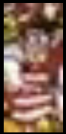


Feature Engineering: Template Matching

Where's Waldo?



Detected Template



Template

Source: GaTech computer Vision

Feature Engineering: Template Matching

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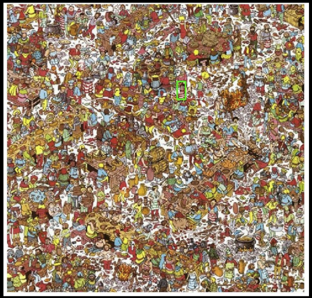

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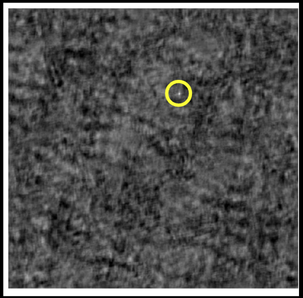

Correlation map

Source: GaTech Computer Vision

TEMPLATE MATCHING - CROSS CORRELATION